

2	(a)	State the principle of <i>conservation of linear momentum</i> .	
			[1]
		The total linear momentum of a system will remain constant if no net external force acts on it.	1

	(b) In a nuclear reactor, carbon atoms are used to slow down neutrons. A fast neutron collides head-on with a stationary carbon atom.
	(i) Show that the impulse acted on the neutron is proportional to the final velocity of the carbon atom in such a collision.
	By conservation of momentum, $m_n u_n + m_c u_c = m_n v_n + m_c v_c$ $m_n v_n - m_n u_n = m_c u_c - m_c v_c$ impulse on neutron, $\Delta p_n = 0 - m_c v_c$ Since the mass of the carbon atom is constant, $\Delta p_n \propto v_c$ OR By conservation of momentum, Total initial momentum of neutron and carbon atom = Total final momentum of neutron and carbon atom Change in momentum of the neutron = - Change in momentum of the carbon atom From definition of impulse, Impulse on the neutron = Change in momentum of the neutron Thus, Impulse on the neutron = - Change in momentum of the carbon atom = - (mass of carbon atom) x (final velocity of carbon atom – initial velocity of carbon atom) = - (mass of carbon atom) x (final velocity of carbon atom) , since carbon atom was initially stationary Since the mass of the carbon atom remains constant, Impulse on the neutron is proportional to the final velocity of the carbon atom.
	(ii) In the collision between a neutron and a carbon atom, a neutron of mass $1.0m$ with initial velocity u collides elastically head-on with a stationary carbon atom of mass $12m$. The final velocities of the neutron and the carbon atom are v and V respectively. By considering the relative speeds between the neutron and carbon atom before and after their collision, show that the fraction of the kinetic energy that is retained by the neutron after such a collision is 0.72.
	$\frac{KE \text{ retained}}{\text{initial KE}} = \frac{KE_{f,n}}{KE_{i,n}}$ $= \frac{\frac{1}{2}mv^2}{\frac{1}{2}mu^2}$ $= \frac{v^2}{u^2} \quad \text{--- (1)}$

		Using Relative speed of approach = Relative speed of separation $u - 0 = V - v$ $V = v + u$ And Conservation of momentum, $m_n u_n + m_c u_c = m_n v_n + m_c v_c$ $mu + 0 = mv + 12mV$ $u = v + 12V$ Therefore, $u = v + 12(v + u)$ $v = -\frac{11}{13}u \text{ --- (2)}$ Sub (2) into (1): $\frac{KE \text{ retained}}{\text{initial KE}} = \frac{\left(-\frac{11}{13}u\right)^2}{u^2}$ $= 0.72$	1
			1
(c)	(i)	Explain why nuclei which are much more massive than carbon atoms are ineffective in slowing down neutrons in the nuclear reactor.	
			
			
			[2]
		<u>Method 1: From working in (b)</u> $\frac{KE \text{ retained}}{\text{initial KE}} = \frac{m_{\text{particle}} - m_n}{m_{\text{particle}} + m_n}$ When $m_{\text{particle}} \gg m_n$, ratio $\frac{KE \text{ retained}}{\text{initial KE}} \approx \frac{m_{\text{particle}}}{m_{\text{particle}}} = 1$ which is the maximum. Thus, when very massive particles are used ($m_{\text{particle}} \gg m_n$), the kinetic energy retained by the neutron will be very large. Derivation in detail (optional): Using Relative speed of approach = Relative speed of separation $V = v + u$ And Conservation of momentum, $m_n u + 0 = m_n v + m_{\text{particle}} V$ $u = v + \frac{m_{\text{particle}}}{m_n} V$ Therefore, $u = v + \frac{m_{\text{particle}}}{m_n} (v + u)$ $v = -\frac{m_{\text{particle}} - m_n}{m_{\text{particle}} + m_n} u \text{ --- (2)}$ Sub (2) into (1):	1
			1
			[1]
			[1]

		$\frac{KE \text{ retained}}{\text{initial } KE} = \frac{\left(-\frac{m_{\text{particle}} - m_n}{m_{\text{particle}} + m_n}u\right)^2}{u^2} = \frac{m_{\text{particle}} - m_n}{m_{\text{particle}} + m_n}$ <u>Method 2: From conservation of momentum</u> When very massive particles are used, by the principle of conservation of momentum, the neutrons will retain most of the magnitude of its initial momentum after collision, traveling in the opposite direction to its initial momentum.	
	(ii)	Explain why particles of similar mass to neutrons such as hydrogen nuclei are not suitable for slowing down neutrons in the nuclear reactor.	
			
			[1]
		<u>Method 1: From working in (b)</u> $\frac{KE \text{ retained}}{\text{initial } KE} = \frac{m_{\text{particle}} - m_n}{m_{\text{particle}} + m_n}$ When $m_{\text{particle}} = m_n$, ratio $\frac{KE \text{ retained}}{\text{initial } KE} = 0$. Thus, when particles of similar mass to neutrons are used ($m_{\text{particle}} = m_n$), the neutrons will stop moving completely which is not the aim (we want the neutrons to move more slowly but not to stop moving otherwise they cannot collide with the fissile isotope). <u>Method 2: From conservation of momentum</u> By the principle of conservation of momentum, when a neutron collides with an <u>initially stationary</u> particle of similar mass (e.g. hydrogen atom), the neutrons will stop moving completely after the collision which is not the aim (we want the neutrons to move more slowly but not to stop moving otherwise they cannot collide with the fissile isotope).	1

3 A light helical spring is suspended vertically from a fixed point as shown in Fig. 3.1